# Han-Bo-Ram Lee

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## Research Interests

Prof. Lee's current research focuses on bridging AI-driven materials synthesis design with experimental realization by elucidating surface chemical reaction mechanisms for next-generation semiconductor device fabrication. His group actively explores a wide range of theoretical methodologies—including deep learning diffusion models, density functional theory combined with machine-learning force fields, and kinetic Monte Carlo simulations—as well as atomic-scale synthesis techniques such as atomic layer deposition (ALD), atomic layer modulation (ALM), and area-selective deposition (ASD). The overarching goal of Prof. Lee's research program is to design novel materials with targeted properties using AI-based approaches and to realize them through advanced silicon device fabrication processes.

## Highlights

Ph.D at POSTECH, Republic of Korea ('09), Postdoc at Stanford, USA ('10-'13), Assistant Prof. ('13-'17) & Associate Professor ('17-'22) & Professor ('22-'26.8) at Incheon National University, Republic of Korea

147 published papers, 11833 citations, 58 h-index, 43 granted patents, 13 pending patents

Executive Editor of Chemistry of Materials (Sep '25-present) and Associate Editor (Jan '18-Aug '25)

Google Scholar · ORCID

## Bio & Research Backgrounds (clickable links below)

Education
Academic & Professional Appointments
Research Experiences
Research Group
Technical Skills

## Research Achievements (clickable links below)

Journal Articles
Book Chapters & Proceedings
Invited Lectures
Press
Patents
Technology Transfer
Invited Conference Presentations
Fellowship and Honor

## Activities (clickable links below)

Teaching Activities
Professional Activities
University and Departmental Committees

## Education

- Ph.D. in Materials Science and Engineering, POSTECH, Republic of Korea
  - Period: 3/2005 - 8/2009
  - Advisor: Professor Hyungjun Kim
  - GPA: 3.7/4.3
  - Thesis: Atomic layer deposition of cobalt
- B.S. in Materials Science and Engineering, Sungkyunkwan University, Republic of Korea
  - Period: 3/1998 - 2/2005 (3/1999 - 5/2002: Military service at the Korean Army)
  - Advisor: Professor Jae Chan Lee
  - GPA: 4.24/4.5, Magna cum laude
  - Thesis: Atomic layer deposition and applications to area-selective growth

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## Academic & Professional Appointments

- Professor, Graduate School of Semiconductor Materials & Devices Engineering, Ulsan National Institute of Science and Technology (UNIST), Republic of Korea
  - Period: Starting 9/2026
- Professor, Department of Materials Science & Engineering, Incheon National University, Republic of Korea
  - Period: 2/2022 - 8/2026
- Executive Editor, Chemistry of Materials, American Chemical Society Publications, USA
  - Period: 9/2025 - present
- Associate Editor, Chemistry of Materials, American Chemical Society Publications, USA
  - Period: 1/2018 - 8/2025
- Consulting Professor, SK hynix, Republic of Korea
  - Period: Starting 3/2026
- Visiting Professor, Department of Chemical Engineering, Stanford University, USA
  - Period: 2/2024 - 3/2026
- Consulting Professor, SK hynix, Republic of Korea
  - Period: 3/2021 - 2/2022
- Visiting Associate Professor, Department of Chemical Engineering, Stanford University, USA
  - Period: 12/2019 - 2/2021

- Associate Professor, Department of Materials Science & Engineering, Incheon National University, Republic of Korea
  - Period: 3/2017 - 2/2022
- Assistant Professor, Department of Materials Science & Engineering, Incheon National University, Republic of Korea
  - Period: 2/2013 - 2/2017
- Postdoctoral Scholar, Department of Chemical Engineering, Stanford University, USA
  - Period: 3/2010 - 1/2013
- Postdoctoral Scholar, Department of Electrical and Electronic Engineering, Yonsei University, Republic of Korea
  - Period: 9/2009 - 2/2010

## Research Experiences

- Postdoctoral Scholar, Department of Chemical Engineering, Stanford University
  - Period: 3/2010 - 1/2013
  - Fund: Department of Energy
  - PI: Professor Stacey F. Bent
  - Research area: Ultra thin atomic layer deposition Pt film for fuel cell (w/ National Renewable Energy Laboratory)
- Postdoctoral Scholar, School of Electrical and Electronic Engineering, Yonsei University
  - Period: 9/2009 - 2/2010
  - Fund: BK21 program
  - PI: Professor Hyungjun Kim
  - Research area: Atomic layer deposition of Co and Ni for metal silicides
- Research Assistant, Dept. of MSE, POSTECH
  - Period: 3/2005 - 8/2009
  - Academic advisor: Professor Hyungjun Kim
  - Research area: Atomic layer deposition of Ru for gate electrode; Atomic layer deposition of Co and Ni for metal silicide contacts; Supercritical fluid deposition of Ru and $SiO_2$
- Visiting Researcher, Department of Biomedical Engineering, UC Irvine.
  - Period: 2/2007 - 2/2007
  - PI: Professor Noo Li Jeon (Jeon's Microfluidic Lab.)
  - Research area: Bio applications of Cobalt nanostructures
- Visiting Researcher, SNTEK Co. Ltd., Kyunggido, Republic of Korea.
  - Period: 1/2005 - 2/2005
  - PI: Kyung Joon Ahn, CEO
  - Research area: Development and fabrication of atomic layer deposition-magnetron sputtering cluster systems
- Undergraduate Researcher, Dept. of MSE, Sungkyunkwan University
  - Period: 9/2004 - 11/2004
  - PI: Professor Jae Chan Lee (Semiconductor Thin Film Devices Lab.)
  - Research area: Atomic layer deposition and application to area-selective growth.

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## Technical Skills

- Thin film deposition equipment
  - Extensive experience with the design, construction, and film growth in atomic layer deposition
  - Nanoscale film depositions by sputtering and evaporation
  - Surface treatment and modification by using plasma or coating monolayer
  - Handling of toxic gases and safety system (design and construction of scrubber and gas lines)
- Thin film analysis
  - TEM sample preparation and imaging technique with energy dispersive spectroscopy and electron energy loss spectroscopy analyses
  - X-ray diffraction, synchrotron radiation XRD and XRR with simulation
  - Scanning electron microscopy with energy dispersive spectroscopy
  - Design and construction of in situ surface analysis equipments using quadrupole mass spectroscopy and synchrotron radiation X-ray reflectivity
  - Atomic force microscopy and roughness analyses
  - Secondary ion mass spectroscopy for depth profile analysis
  - Rutherford backscattering and simulation for depth profile analysis
  - X-ray photoelectron spectroscopy for chemical qualification and quantification analyses
  - Auger electron spectroscopy for depth profile and area mapping analyses
  - Glow discharge optical emission spectroscopy for depth profile analysis
  - Surface contact angle measurement for hydrophobicity analysis
  - Inductively-coupled plasma mass spectroscopy for metal quantification
- Device fabrication
  - Rapid thermal annealing (forming gas annealing, oxidation, reduction process)
  - Photolithography
  - Wet chemical etching and reactive ion etching
- Electrical property measurement:
  - Capacitance-voltage (CV) and current-voltage (IV) measurements for metal-oxide-semiconductor characterizations
  - 4-point probe measurement for thin film resistivity
- Software tools:
  - Mathematica for calculation and fitting data
  - Nanoparticle analysis using computational method (ImageJ, Gatan Microscope)

- LabVIEW for control of equipment and data acquisition (Certificates of “LabVIEW Basics II Course” and “Signal Conditioning Course” from National Instrument)
- SPEC and C-PLOT for control of X-ray goniometer and data acquisition of X-ray diffraction
- Parratt32 for X-ray reflectivity simulation, RUMP for Rutherford backscattering simulation

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# Research Group

CURRENT MEMBERS

### Postdoctoral Scholars

- Dr. Aravind H. Patil, Postdoctoral Scholar (Korea Polar Research Institute (KOPRI) & Dept. of Materials Science Engineering, INU, January 2022 – August 2026 → KOPRI & Graduate School of Semiconductor Materials & Devices Engineering, UNIST, September 2026 – present)
  - Superhydrophobic/Icephobic coatings, Atomic layer deposition
- Dr. Summal Zoha, Postdoctoral Scholar (Dept. of Materials Science Engineering, INU, June 2026 – present)
  - Area selective ALD
- Dr. Ngoc Le Trinh, Postdoctoral Scholar (Dept. of Materials Science Engineering, INU, June 2026 – August 2026 → Graduate School of Semiconductor Materials & Devices Engineering, UNIST, September 2026 – present)
  - Atomic layer modulation, Density functional theory

### Ph.D. Students

- Zunair Masroor, Ph.D. Course (Dept. of Materials Science Engineering, INU, September 2022 – present)
  - Molecular dynamics, Deep learning, Density functional theory, AI-driven ALD
- Kyeongmin Min, Ph.D. Course (Dept. of Materials Science Engineering, INU, March 2025 – August 2026 → Graduate School of Semiconductor Materials & Devices Engineering, UNIST, September 2026 – present)
  - Hot wire ALD, Area selective ALD
- Byungchan Lee, Ph.D. Course (Dept. of Materials Science Engineering, INU, June 2025 – August 2026 → Graduate School of Semiconductor Materials & Devices Engineering, UNIST, September 2026 – present)
  - Area selective ALD, AI-driven ALD, Deep learning
- Hamna Fatima, Ph.D. Course (Dept. of Materials Science Engineering, INU, September 2024 – present)
  - Area selective ALE

- Wonjoong Kim, Ph.D. Course (Dept. of Intelligent Semiconductor Engineering, INU, March 2026 – August 2026 → Graduate School of Semiconductor Materials & Devices Engineering, UNIST, September 2026 – present)
  - Atomic layer modulation, Kinetic MC simulation, Density functional theory, AI-driven ALD

### MS Students

- Yoonseo Choi, MS Course (Dept. of Intelligent Semiconductor Engineering, INU, March 2025 – August 2026 → Graduate School of Semiconductor Materials & Devices Engineering, UNIST, September 2026 – present)
  - TiN ALD reactant chemistry, Atomic layer modulation of metal alloys
- Kwangyong An, MS Course (Dept. of Materials Science Engineering, INU, August 2025 – August 2026 → Graduate School of Semiconductor Materials & Devices Engineering, UNIST, September 2026 – present)
  - Mo ALD, InN ALD
- Minhyeok Lee, MS Course (Dept. of Intelligent Semiconductor Engineering, INU, March 2025 – present)
  - Kinetic MC simulation, AI-driven ALD
- Dohyun Kim, MS Course (Dept. of Intelligent Semiconductor Engineering, INU, March 2026 – August 2026 → Graduate School of Semiconductor Materials & Devices Engineering, UNIST, September 2026 – present)
  - Kinetic MC simulation, AI-driven ALD

### Undergraduate Internship

- Nayoung Lee, Undergraduate Internship (Dept. of Materials Science Engineering, INU, June 2025 – present)
  - High crystallinity in ultrathin films

## ALUMNI

### Ph.D. Students

- Dr. Bonwook Gu (now at SK hynix), Ph.D. Course (Dept. of Materials Science Engineering, INU, March 2022 – July 2026), Metal ALD, kinetic MC simulation, AI-driven ALD
- Dr. Chi Thang Nguyen (now at IFW Dresden), Ph.D. Course (Dept. of Materials Science Engineering, INU, March 2018 – August 2023), Area selective ALD, $TiO_2$
- Dr. Sumaira Yasmeen (now at University of Colorado at Boulder), Ph.D. Course (Dept. of Materials Science Engineering, INU, September 2018 – September 2022), Selective deposition of Ru and $SiO_2$

- Dr. Hyun Gu Kim (now at Wonik IPS), Ph.D. Course (Dept. of Materials Science Engineering, INU, March 2014 – August 2019), Area selective ALD, ALD of Pt on reduced graphene oxide
- Dr. Jong Seo Park, Ph.D. Course (Co-advising with Prof. Hyungjun Kim at Yonsei University, September 2015 – August 2019), $HfO_2$ ALD for ferroelectricity

### MS Students

- Hye Won Park, MS Course (Dept. of Materials Science Engineering, INU, March 2024 – December 2025), TiN ALD
- Dabin Gong (now at ASM Korea), BS/MS Co-terminal Course (Dept. of Materials Science Engineering, INU, September 2022 – February 2025), ALD, AS-ALD
- Min Kyu Lee (now at Samsung Electronics), MS Course (Dept. of Materials Science Engineering, INU, March 2022 – August 2024), ALD, ASD
- Chan Hui Mun (now at Samsung Electronics), BS/MS Co-terminal Course (Dept. of Materials Science Engineering, INU, March 2020 – February 2022), Superhydrophobic coating, icephobic coating
- Byung Kook Ko (now at Samsung Electronics), MS Course (Dept. of Materials Science Engineering, INU, March 2019 – February 2021), Hf-based ferroelectric materials, atomic layer modulation

### Postdoctoral Scholars

- Dr. Abu Saad Ansari, Research Professor (Dept. of Materials Science Engineering, INU, March 2021 – October 2021), DFT, Area selective ALD
- Dr. Rizwan Khan, Postdoctoral Scholar (Dept. of Materials Science Engineering, INU, March 2017 – May 2021), Area selective ALD
- Dr. Houda Gaiji, Postdoctoral Scholar (Dept. of Materials Science Engineering, INU, July 2019 – May 2020), Superhydrophobic coating for icephobic property
- Dr. Jae Hong Yun, Postdoctoral Scholar (Dept. of Materials Science Engineering, INU, March 2019 – October 2019), Ru ALD, superhydrophobic coating

### Undergraduate Internship

- You Chan Ko, Undergraduate Internship (Dept. of Materials Science Engineering, INU, June 2025 – May 2026), Hot wire ALD, ASD
- Min Sung Kim (now at Applied Materials Korea), Undergraduate Internship (Dept. of Materials Science Engineering, INU, February 2020 – November 2022), MC Simulation for precursor adsorption

- Hyuna Bang, Undergraduate Internship (Dept. of Materials Science Engineering, INU, December 2021 – March 2022), ALD, AS-ALD
- Soohyun Choi, Undergraduate Internship (Dept. of Materials Science Engineering, INU, February 2021 – August 2021), Area selective ALD
- Min Hee Shong, Undergraduate Internship (Dept. of Materials Science Engineering, INU, March 2021 – June 2021), Hf-based ferroelectric materials, atomic layer modulation
- A Young Lim, Undergraduate Internship (Dept. of Materials Science Engineering, INU, February 2020 – December 2020), MC Simulation for precursor adsorption
- Joonhyuk Ko, Undergraduate Internship (Dept. of Materials Science Engineering, INU, July 2020 – February 2021), Superhydrophobic coating
- Hyeonseok Oh, Undergraduate Internship (Dept. of Materials Science Engineering, INU, July 2020 – February 2021), Atomic layer modulation, kinetic MC simulation
- Geonwoo Park, Undergraduate Internship (Dept. of Materials Science Engineering, INU, July 2020 – February 2021), MC Simulation for precursor adsorption
- Seokjun Seo, Undergraduate Internship (Dept. of Materials Science Engineering, INU, July 2020 – January 2021), Superhydrophobic coating
- Jinsic Kim, Undergraduate Internship (Dept. of Materials Science Engineering, INU, March 2021 – January 2021), Area selective ALD
- Sang Hui Han, Undergraduate Internship (Dept. of Materials Science Engineering, INU, November 2019 – June 2020), Area selective ALD
- Jongwoo Shin, Undergraduate Internship (Dept. of Materials Science Engineering, INU, March 2019 – February 2020), Selective deposition by ALD
- Je Young Ha, Undergraduate Internship (Dept. of Materials Science Engineering, INU, September 2019 – February 2020), Area selective ALD
- Se Hyun Shong, Undergraduate Internship (Dept. of Materials Science Engineering, INU, September 2019 – February 2020), Area selective ALD
- Ho Yeon Kim, Undergraduate Internship (Dept. of Materials Science Engineering, INU, December 2018 – August 2019), Area selective ALD
- Kyung Taek Park, Undergraduate Internship (Dept. of Materials Science Engineering, INU, December 2018 – August 2019), Area selective ALD
- Woo Hyuk Kwon, Undergraduate Internship (Dept. of Materials Science Engineering, INU, December 2016 – December 2018), Electronic textiles, area selective ALD
- Jae Kwang Lee, Undergraduate Internship (Dept. of Materials Science Engineering, INU, June 2017 – December 2018), Superhydrophobic coating, area selective ALD

- Tae Hwan Im, Undergraduate Internship (Dept. of Materials Science Engineering, INU, October 2017 – November 2018), Area selective ALD
- Hye Jin Lee, Undergraduate Internship (Dept. of Materials Science Engineering, INU, December 2017 – November 2018), Superhydrophobic coating
- Jae Hee Jung, Undergraduate Internship (Dept. of Materials Science Engineering, INU, January 2018 – August 2018), Area selective atomic layer deposition
- Seung Jae Ahn, Undergraduate Internship (Dept. of Materials Science Engineering, INU, January 2018 – November 2018), Area selective atomic layer deposition
- Yong Woon Lee, Undergraduate Internship (Dept. of Materials Science Engineering, INU, June 2017 – April 2018), Ferroelectric tunnel junction by ALD
- Yung Hak Lee, Undergraduate Internship (Dept. of Materials Science Engineering, INU, December 2016 – February 2018), Area selective ALD
- Tae Hee Han, Undergraduate Internship (Dept. of Materials Science Engineering, INU, December 2016 – August 2017), Electronic textiles, smart textile heater
- Sun Hwa Kim, Undergraduate Internship (Dept. of Materials Science Engineering, INU, September 2015 – December 2016), Chemical bath deposition of rare earth oxides for hydrophobic coating
- Tae Hoon Park, Undergraduate Internship (Dept. of Materials Science Engineering, INU, September 2014 – December 2015), ALD of Pt on carbon powder, ALD of Pt on reduced graphene oxide
- Min Ji Kim, Undergraduate Internship (Dept. of Materials Science Engineering, INU, September 2014 – December 2015), ALD of $TiO_2$ for photocatalyst

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## Journal Articles

SCI & SCIE (147 PUBLISHED, 6 SUBMITTED, 11833 CITATIONS, 58 H-INDEX)

1. "Experimentally Grounded Machine Learning of Precursor–Surface–Reactant Chemistry in Atomic Layer Deposition," Bonwook Gu, Ngoc Le Trinh, Zunair Masroor, Minhyeok Lee, and Han-Bo-Ram Lee*, submitted to npj Computational Materials, [Corresponding author]
2. "Dual-Function Aminosilane and Ozone Chemistry for Integrated Area-Selective $SiO_2$ Deposition and Area-Selective Ru Etching," Summal Zoha, Sumaira Yasmeen, Chi Thang Nguyen, Zunair Masroor, Bonwook Gu, Youngho Kang, and Han-Bo-Ram Lee*, submitted to ACS Applied Nano Materials, [Corresponding author]
3. "Oxidant-Driven Modulation of Interfacial Oxidation and Electrical Properties of $HfO_2$ on TiN by ALD," Aravind H. Patil†, Byungchan Lee†, Sumin An, Seoung Won Shim, Wonjoong Kim, Minhyeok Lee, Sanh Vo Thi, Joondong Kim, Youngho Kang, and Han-Bo-Ram Lee*, submitted to Journal of Materials Chemistry C, †Equal contribution, [Corresponding author]
4. "Atomic-Layer Sequence Engineering for Thermally Robust Ultrathin High-k Dielectrics," Byung-Ha Kwak, Ngoc Le Trinh, Yea-Ji Kim, Wonjoong Kim, Hyun-Mi Kim, Han-Bo-Ram Lee, and Il-Kwon Oh*, submitted to Applied Surface Science, [Co-author]
5. "Orientation-controlled Area-selective Deposition of Near-single-grain Ruthenium Interconnects," Eun-Hyoung Cho*, Dongmin Kim, Iaan Cho, Bonggeun Shong, Kyeongmin Min, and Han-Bo-Ram Lee, submitted to Nature Nanotechnology, [Co-author]
6. "Deep-Learning-Accelerated Dopant Selection for High-k $HfO_2$ Dielectrics: A Disorder-Resolved Study of Y, Si, and Al," Zunair Masroor, Bonwook Gu, Wonjoong Kim, Ngoc Le Trinh, Summal Zoha, and Han-Bo-Ram Lee*, Journal of Physical Chemistry C, online published, 10/23/2026, https://doi.org/10.1021/acs.jpcc.6c05599, [Corresponding author]
7. "Transparent and Photovoltaic Water–Semiconductor Interface for Ultra-Low-Light Omnidirectional Photodetection," Sanh Vo Thi, Malkeshkumar Patel, Sourov Hossain, Md Arifur Rahman Barno, Ngoc Le Trinh, Han-Bo-Ram Lee, Yu Kwon Kim*, and Joondong Kim*, Energy & Environmental Materials, e70500, online published, 8/18/2026, https://doi.org/10.1002/eem2.70500, [Co-author]
8. "Synergistic Reinforcement between Al and Si," Han-Bo-Ram Lee*, Chemistry of Materials, 38 (14), 7043-7046 (2026), 7/28/2026, https://doi.org/10.1021/acs.chemmater.6c01759, [Corresponding author]

9. “Chemically Distinct Adsorption of Sulfide Inhibitors for Multi-Surface Passivation and Area Selective Deposition,” Summal Zoha†, Fabian Pieck†, Bonwook Gu, Zunair Masroor, Ralf Tonner-Zech*, and Han-Bo-Ram Lee*, Small, e74721, online published, 7/19/2026, https://doi.org/10.1002/smll.74721, †Equal contribution, [Corresponding author]
10. “Atomic Layer Deposition of Titanium Nitride: Reactant-Dependent Ligand Removal and Oxygen Incorporation,” Yoonseo Choi, Byungchan Lee, Ngoc Le Trinh, Hyewon Park, and Han-Bo-Ram Lee*, submitted to Journal of Physical Chemistry C, [Corresponding author]
11. “Mechanistic Atomic Hydrogen Chemistry for Ruthenium Deposition: From Ligand Elimination to Area-Selective Patterning,” Kyeongmin Min, Chi Thang Nguyen, Eun-Hyoung Cho, Minhyeok Lee, Wonjoong Kim, Youngmin Sunwoo, Eunhyeok Heo, Byungjo Kim, Hao Van Bui and Han-Bo-Ram Lee*, Journal of the American Chemical Society, 148 (27), 28484-28497 (2026), 6/9/2026, https://doi.org/10.1021/jacs.6c04779, [Corresponding author]
12. “Overcoming the Critical Thickness Limit: Interfacial Control of Crystallization Pathways in Atomic-Scale Dielectric Thin Films,” Ngoc Le Trinh, Wonjoong Kim, Minhyeok Lee, Bonwook Gu, Sanh Vo Thi, Ji Liu, Michael Nolan, Hyun-Mi Kim, Hyeongkeun Kim, Joondong Kim, Youngho Kang, and Han-Bo-Ram Lee*, ACS Nano, 20 (24), 17273-17285 (2026), 6/8/2026, https://doi.org/10.1021/acsnano.6c00649, [Corresponding author]
13. “Defect Engineering and Improving Dielectric Performance of Y-Doped $ZrO_2$ Ultrathin Film through Atomic Layer Modulation,” Ngoc Le Trinh, Wonjoong Kim, Minhyeok Lee, Nayoung Lee, Zunair Masroor, Byung-Ha Kwak, Hyun-Mi Kim, Hyeongkeun Kim, Il-Kwon Oh, and Han-Bo-Ram Lee*, Applied Surface Science, 753, 168428 (2026), 9/2026, https://doi.org/10.1016/j.apsusc.2026.168428, [Corresponding author]
14. “A Dual-Si Atomic Precursor Enabling High-Quality $SiN_x$ Thin Films via Very High-Frequency Plasma Atomic Layer Deposition,” Young-Jin Lim, Yujin Lee, Kim-Hue Thi Dinh, Min-Jeong Rhee, Yea-Ji Kim, Ngoc Le Trinh, Bonwook Gu, Youngho Kang, Jae Hack Jeong, Han-Bo-Ram Lee*, Stacey Bent*, and Il-Kwon Oh*, Chemistry of Materials, 38 (11), 5679–5690 (2026), 5/16/2026, https://doi.org/10.1021/acs.chemmater.6c00586, [Corresponding author]
15. “AI-Driven Inverse Design of Complex Oxide Thin Films for Semiconductor Devices: A Case Study of Hf–Zr–O,” Bonwook Gu, Ngoc Le Trinh, Wonjoong Kim, Zunair Masroor, and Han-Bo-Ram Lee*, Chemistry of Materials, 38 (10), 5285–5297 (2026), 5/11/2026, https://doi.org/10.1021/acs.chemmater.6c00978, [Corresponding author]
16. “Atomic Layer Deposition of Molybdenum Carbide and Substrate-dependent Reduction to Metallic Molybdenum,” Kwangyong An, Kieran G. Lawford, Bonwook Gu, Aravind H. Patil, Aaron

D. Rogers, Seán T. Barry, and Han-Bo-Ram Lee*, Journal of Materials Chemistry C, 14 (23), 10006-10019 (2026), 4/14/2026, https://doi.org/10.1039/D6TC00454G, [Corresponding author]

17. "Solar-Driven Icephobic Layer Design by Superhydrophobic Materials," Aravind H. Patil, Sumin An, Youngho Kang, Sawanta S. Mali, Chang Kook Hong, Joohan Lee, Changhyun Chung*, and Han-Bo-Ram-Lee*, Applied Surface Science, 732, 166580 (2026), 3/13/2026, https://doi.org/10.1016/j.apsusc.2026.166580, [Corresponding author]
18. "Tuning Surface Reactivity Pathways through Molecular Inhibitor Redosing for Precision Nanopatterning," Byungchan Lee, Chi Thang Nguyen, Minhyeok Lee, Ngoc Le Trinh, Kyeongmin Min, Younho Kang, Eun-Hyoung Cho, and Han-Bo-Ram Lee, Materials Horizon, 13 (10), 5035-5049 (2026), 3/3/2026, https://doi.org/10.1039/D5MH02445E, [Corresponding author]
19. "Cobalt Precursor as Surface Inhibitor for Area-Selective Atomic Layer Deposition of Nanoscale Electronic Interconnects," Zunair Masroor, Ngoc Le Trinh, Summal Zoha, Youngho Kang, and Han-Bo-Ram Lee*, ACS Applied Nano Materials, 9 (4), 1869-1878 (2026), 1/21/2026, https://doi.org/10.1021/acsanm.5c05042, [Corresponding author]
20. "Atomic Layer Modulation of Ruthenium Aluminum Oxide through Reactivity Control of Precursors for Seedless Copper Interconnects," Chi Thang Nguyen, Miso Kim, Youn-Hye Kim, Taehoon Cheon, Ngoc Le Trinh, Sehee Kim, Soo-Hyun Kim, Bonggeun Shong*, and Han-Bo-Ram Lee*, Chemistry of Materials, 37 (24), 9745-9757 (2025), 11/27/2025, https://dx.doi.org/10.1021/acs.chemmater.5c02156, [Corresponding author]
21. "Area Selective Deposition of Tungsten and Ruthenium using a Tungsten Precursor Inhibitor," Bonwook Gu, Chi Thang Nguyen, Mingyu Lee, Sunghwan Jo, Ngoc Le Trinh, Youngho Kang, Seung Wook Ryu and Han-Bo-Ram Lee*, Applied Surface Science, 720, 165129 (2026), 11/6/2025, https://doi.org/10.1016/j.apsusc.2025.165129, [Corresponding author]
22. "Materials Chemistry in the Far South," Han-Bo-Ram Lee*, 37 (20), 8103-8106 (2025), Chemistry of Materials, http://doi.org/10.1021/acs.chemmater.5c02391, [Corresponding author]
23. "Functionalization of $Si_3N_4$ with Aldehyde Inhibitor for Area-Selective Deposition of $HfO_2$ and $SiO_2$," Byung Chan Lee, Seoung Won Shim, Ngoc Le Trinh, Aravind H. Patil, Chi Thang Nguyen, Taehoon Cheon, Soo-Hyun Kim, Youngho Kang, and Han-Bo-Ram Lee*, Journal of Materials Chemistry C, 13 (37), 19316-19329 (2025), 8/11/2025, https://doi.org/10.1039/D5TC01969A, [Corresponding author]
24. "Ultrathin Ag Film of Transparent Electrode for Flexible Schottky Photodetection and Enhanced Charge Transport," Sanh Vo Thi, Malkeshkumar Patel, Thanh Tai Nguyen, Sourov Hossain, Seoyoung Lim, Ngoc Le Trinh, Han-Bo-Ram Lee, Dong-Wook Kim*, and Joondong Kim*, ACS

Applied Electronic Materials, 7(9), 4339-4351 (2025), 4/8/2025, https://doi.org/10.1021/acsaelm.5c00498, [Co-author]

25. "Patterning Technology: From Supporting Role to Main Player in Materials Chemistry," Han-Bo-Ram Lee*, Chemistry of Materials, 37 (6), 2071-2072 (2025), 3/25/2025, https://doi.org/10.1021/acs.chemmater.5c00419, [Corresponding author]
26. "Synergistic Effects of Ionic Liquid Integrated Superhydrophobic Composite Coatings on Anti-Icing Properties," Aravind H. Patil, Ngoc Le Trinh, Hackwon Do, Youngho Kang, Joohan Lee, Changhyun Chung*, and Han-Bo-Ram Lee*, 693, 162749 (2025), 2/21/2025, Applied Surface Science, https://doi.org/10.1016/j.apsusc.2025.162749, [Corresponding author]
27. "Atomic-level Stoichiometry Control of Ferroelectric $Hf_xZr_yO_z$ Thin Films by Understanding Molecular-level Chemical Physical Reactions," Ngoc Le Trinh, Bonwook Gu, Kun Yang, Chi Thang Nguyen, Byungchan Lee, Hyun-Mi Kim, Hyeongkeun Kim, Youngho Kang, Min Hyuk Park*, and Han-Bo-Ram Lee*, ACS Nano, 19(3), 3562-3578 (2025), 1/15/2025, https://doi.org/10.1021/acsnano.4c13595, [Corresponding author]
28. "Insights into the Thermal Decomposition Mechanism of Molybdenum(VI) Iminopyridine Adducts," Lara K. Watanabe, Aaron D. Rogers, Marshall T. Atherton, Kieran G. Lawford, Bonwook Gu, Supill Chun, Sanghyun Lee, Dongkyun Lee, Han-Bo-Ram Lee, and Seán T. Barry, Chemistry of Materials, 36(24), 11985-11995 (2024), 12/5/2024, https://doi.org/10.1021/acs.chemmater.4c02689, [Co-author]
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21. Han-Bo-Ram Lee, Hyungjun Kim, Bo-Eun Park, Il-Kwon Oh, “Enhanced electrical characteristic Electric Device for MIM Capacitor and the Manufacturing Method thereof,” KR 10-1752060, 6/22/2017
22. Han-Bo-Ram Lee, Hyungjun Kim, Jun Ho Choi, Il-Kwon Oh, “Method for Modifying Surface of Substrate Using Rare Earth Oxide Thin Film,” KR 10-1751619, 6/21/2017
23. Taeyoon Lee, Jong Hyun Ahn, Juree Hong, Jae-Bok Lee, Han-Bo-Ram Lee, “Method for Healing Defect of Conductive Layer, Method for Forming Metal-carbon Compound Layer, 2D Nano Materials, Transparent Electrode and Method for Manufacturing the Same,” KR 10-1720168, 3/21/2017
24. Han-Bo-Ram Lee, “3D Conductive Coating Method For Clear SEM Measurement,” KR 10-1686386, 12/7/2016
25. Hyungjun Kim, Il-Kwon Oh, Chang-Mo Yun, Han-Bo-Ram Lee, “Filter For Removing Moisture And Method For Manufacturing The Same,” KR 10-1670337, 10/24/2016
26. Han-Bo-Ram Lee, Hyun Gu Kim, “Manufacturing Method of Ultrathin Continuous Metal Film Using Surface Funtionalization,” KR 10-1672984, 10/31/2016

27. Hyungjun Kim, Il-Kwon Oh, Han-Bo-Ram Lee, “Apparatus for Plasma Enhanced Atomic Layer Deposition and Method for Forming Thin Film Oxides Using The Same,” KR 10-1662194, 9/27/2016
28. Hyungjun Kim, Soo Hyun Kim, Jaehong Yun, Han-Bo-Ram Lee, “Semiconductor Device and Method For Manufacturing the Same,” US 9,396,994, 7/19/2016
29. Hyungjun Kim, Soo Hyun Kim, Jaehong Yun, Han-Bo-Ram Lee, “Semiconductor Device and Method For Manufacturing the Same,” KR 10-1621852, 5/11/2016
30. Hyungjun Kim, Il-Kwon Oh, Han-Bo-Ram Lee, “Superhydrophobic Coating Material and Method for Manufacturing the Same,” KR 10-1617396, 4/26/2016
31. Hyungjun Kim, Il-Kwon Oh, Han-Bo-Ram Lee, “Method For Forming Coating Layer and Coating Material Having Waterproof Property,” KR 10-1615897, 4/21/2016
32. Han-Bo-Ram Lee, “Manufacturing Method of Nanowire using Atomic Layer Deposition,” KR 10-1602628, 3/7/2016
33. Gilsang Yoo, Il-Kwon Oh, Hyungjun Kim, Han-Bo-Ram Lee, “Apparatus Method for Performing Plasma Enhanced Atomic Layer Deposition Employing Very High Frequency,” KR 10-1596329, 2/16/2016
34. Han-Bo-Ram Lee and Kwan Pyo Kim, “Healing Method of Defect using Atomic Layer Deposition,” KR 10-1568159, 11/5/2015
35. Jaehong Yun, Soo Hyun Kim, Hyungjun Kim, Han-Bo-Ram Lee, “Copper Layer Capping Method and Method for Fabricating Copper Wire Employing the Same,” KR 10-1553357, 9/9/2015
36. Han-Bo-Ram Lee, Jaemin Kim, Jeong Won Lee, Heung-Soon Lee, Hyungjun Kim, “Method for Fabrication of Metal Nanostructures by using Supercritical Fluid Deposition,” KR 10-1124351, 2/29/2012
37. Han-Bo-Ram Lee, Gil Ho Gu, C.G. Park, Hyungjun Kim, “Manufacturing Method of Metal Nanostructures,” KR 10-1124352, 2/29/2012
38. Han-Bo-Ram Lee, Woo-Hee Kim, Hyungjun Kim, “Method for Forming Contacts of Semiconductor Devices using The Selective Deposition,” KR 10-1078309, 10/25/2011
39. Han-Bo-Ram Lee, Gil Ho Gu, C.G. Park, J.Y. Son, Hyungjun Kim, “Fabrication Method of Catalyst-less Metal Nanorods by Plasma-Enhanced Atomic Layer Deposition and A Semiconductor Element,” KR 10-0920456, 9/29/2009
40. Han-Bo-Ram Lee, J.Y. Son, Hyungjun Kim, “Manufacturing Method of Metal Silicide Thin Layer by Plasma-Enhanced Atomic Layer Deposition without Annealing,” KR 10-0920455, 9/29/2009

41. Han-Bo-Ram Lee, J.Y. Son, Hyungjun Kim, "Manufacturing Method of Metal Silicide by Plasma-Enhanced Atomic Layer Deposition for Contact Application in Semiconductor Devices," KR 10-0872799, 12/2/2008
42. Han-Bo-Ram Lee, Gil Ho Gu, C.G. Park, Hyungjun Kim, "Method for Forming Metal-Silicide Layer in Semiconductor Devices using Plasma Nitridation," KR 10-0872801, 12/2/2008
43. Hyungjun Kim, Y.H. Shin, Y.S. Woo, Han-Bo-Ram Lee, J.Y. Son, "Transistor and Nonvolatile Memory using Deformation Resistivity of Carbon Nanotube and Piezoelectric Effect," KR 10-0848813, 7/28/2007

PENDING

1. Han-Bo-Ram Lee, Hyewon Park, "Atomic Layer Deposition Method and Semiconductor Device using the Same," 10-2025-0029862, 3/7/2025
2. Han-Bo-Ram Lee, Summal Zoha, Yongjin Kim, Sanghoon Ahn, Minkyung Lee, Woojin Lee, Hangyeol Choi, "Semiconductor Devices," 10-2024-0087160, 7/2/2024
3. Han-Bo-Ram Lee, Il-Kwon Oh, "Dysprosium as a Crystalline Aligner in $HfO_2$ Thin Films by Atomic Layer Deposition," 10-2022-0164227, 11/30/2022
4. Han-Bo-Ram Lee, Summal Zoha, "Area Selective Deposition of $HfO_2$ ALD on Metal, Oxide, and Nitride Using Organothiol Inhibitor," 10-2022-0047877, 4/19/2022
5. Han-Bo-Ram Lee, Do Han Lee, Eun Soo Kim, Seung Wook Ryu, Sung Hwan Jo, Abu Saad Aqueel Ahmad Ansari, Ngoc Le Trinh, "Thin Film Deposition Method And Manufacturing Method Of Electronic Device Applying The Same," 10-2022-0021554, 2/18/2022
6. Taekjib Choi, Ho-Jin Lee, Jun-Bong Lee, Han-Bo-Ram Lee, Hong-Woon Lee, Byung-Kook Ko, Jae Kwang Lee, "Self-rectifying Ferroelectric Tunnel Junction Memory Device and Crosspoint Array Having the Same," 10-2018-0037603, 3/30/2018
7. Han-Bo-Ram Lee, "Fabrication Method of High Quality Materials for Quadruple Patterning using Heteroatom Alloying," PCT/KR2017/013109, 11/17/2017
8. Hyungjun Kim, Il-Kwon Oh, Chang-Mo Yun, Han-Bo-Ram Lee, "Method for Forming Thin Film," PCT/ KR2016/014333, 12/7/2016
9. Hyungjun Kim, Il-Kwon Oh, Chang-Mo Yun, Han-Bo-Ram Lee, "Filter For Removing Moisture And Method For Manufacturing The Same," PCT/KR2016/014160, 12/2/2016
10. Taeyoon Lee, Jong Hyun Ahn, Juree Hong, Jae-Bok Lee, Han-Bo-Ram Lee, "Method for Healing Defect of Conductive Layer, Method for Forming Metal-Carbon Compound Layer, 2D Nano

Materials, Transparent Electrode and Method for Manufacturing the Same," US 15/196,820, 06/29/2016

11. Han-Bo-Ram Lee, Hyungjun Kim, Chang Mo Yoon, Il-Kwon Oh, "A Functional Filter and Manufacturing Method of The Same," KR 10-2015-0171229, 12/3/2015
12. Hyungjun Kim, Il-Kwon Oh, Han-Bo-Ram Lee, "Method for Forming Coating Layer and Coating Material Having Waterproof Property," US 14/814,888, 07/31/2015
13. Jaehong Yun, Han-Bo-Ram Lee, Hyungjun Kim, "Method for Fabrication of Silicide Nanowires with Ru Capping Layer," KR 10-2010-0140092, 12/31/2010

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## Technology Transfer

1. "Fabrication Method of Conductive Fibers," KR 10-1846084, transferred to Hanil Corporation, 5/30/2025
2. "Method for Forming Superhydrophobic Coating Layer using PDMS and Micropowders," KR 10-1871073, transferred to Pyocell Co., Ltd., 9/23/2024
3. "Method for Forming Hydrophobic Coating Layer using Structuralization of PDMS Surface," KR 10-1807407, transferred to H-Steel Co., Ltd., 8/9/2024
4. "Manufacturing Method of Ultrathin Continuous Metal Film Using Surface Functionalization," KR 10-1672984, transferred to Moon-Ju Park, 1/4/2024
5. "Healing Method of Defect using Atomic Layer Deposition," KR 10-1568159, transferred to ADWIN Co., Ltd., 11/15/2023
6. "Manufacturing Method of Nanowire using Atomic Layer Deposition," KR 10-1602628, transferred to Daon F&E Co., Ltd., 4/7/2023
7. "3D Conductive Coating Method For Clear SEM Measurement," KR 10-1686386, transferred to Moman Co., Ltd., 8/30/2018

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# Book Chapters & Proceedings

## BOOK CHAPTERS

1. Summal Zoha, Byungchan Lee, and Han-Bo-Ram Lee*, "Defect Chemistry and Associated Organic Functionalization Strategies in Area-Selective Deposition," in the section of "Organic Functionalization of Inorganic Surafces," submitted to the Encyclopedia of Interfacial Chemistry, 2nd Edition, Elsevier. [Corresponding author]
2. Hyungjun Kim, Soo Hyun Kim, and Han-Bo-Ram Lee, "Back End of the Line," in Atomic Layer Deposition for Semiconductors, Ed. Cheol Seong Hwang, Springer, ISBN: 978-1-4614-8054-9, 10/19/2013, http://dx.doi.org/10.1007/978-1-4614-8054-9_8, [Co-author]
3. Han-Bo-Ram Lee and Stacey F. Bent, "Chapter 9. Nanopatterning by Area-Selective Atomic Layer Deposition," in Atomic Layer Deposition of Nanostructured Materials, Ed. Nicola Pinna and Mato Knez, Wiley-VCH (2011), ISBN: 978-3-527-32797-3, 1/2/2012, http://dx.doi.org/10.1002/9783527639915.ch9, [First author]

## PROCEEDINGS

1. "Contact Resistance Reduction using Fermi Level De-pinning Layer for $MoS_2$ FETs," Woojin Park, Yonghun Kim, Sang Kyung Lee, Ukjin Jung, Jin Ho Yang, Chunhum Cho, Yun Ji Kim, Sung Kwan Lim, In Seol Hwang, Han-Bo-Ram Lee, and Byoung Hun Lee, International Electron Device Meeting, 5.1.1 - 5.1.4 (2014), http://dx.doi.org/10.1109/IEDM.2014.7046986, [Co-author]
2. Hyungjun Kim, Jaehong Yoon, and Han-Bo-Ram Lee, "Atomic layer deposition for nanoscale contact applications," Interconnect Technology Conference and 2001 Materials for Advanced Metallization (IITC/ MAM), 2011 IEEE International, 8-12, 1-3(2011), http://ieeexplore.ieee.org/xpls/abs_all.jsp?arnumber=5940260, [Co-author]
3. Hyungjun Kim, Han-Bo-Ram Lee, Woo-Hee Kim, Sang-Joon Park, and In Chan Hwang, "Nanomaterials Fabrication using Advanced Thin Film Deposition and Nanohybrid Process," IEEE Nanotechnology Materials and Devices Conference 2009, 5167578, 3-4(2009), http://ieeexplore.ieee.org/xpls/abs_all.jsp?arnumber=5167578, [Co-author]
4. Hyungjun Kim, Woo-Hee Kim, Han-Bo-Ram Lee, and S. J. Lim, "The Benefits of Atomic Layer Deposition in Non-Semiconductor Applications; Producing Metallic Nanomaterials and Fabrication of Flexible Display," ECS Transactions, 25, 101(2009), http://dx.doi.org/10.1149/1.3205047, [Co-author]

5. Han-Bo-Ram Lee, Kwang Heo, Seunghun Hong, and Hyungjun Kim, “Formation of Silicide Nanowires by Annealing of Atomic Layer Deposition Cobalt/Silicon Core-Shell Nanowires,” ECS Transactions, 25, 175(2009), http://dx.doi.org/10.1149/1.3205052, [First author]
6. (Invited) Han-Bo-Ram Lee and Hyungjun Kim, “Atomic Layer Deposition for Nanoscale Contact Applications,” 25th International VLSI/ULSI Multilevel Interconnection Conference Proceeding, 39(2008), [First author]
7. Han-Bo-Ram Lee and Hyungjun Kim, “Area Selective Atomic Layer Deposition of Cobalt Thin Films”, ECS Transactions, 16, 219(2008), http://dx.doi.org/10.1149/1.2979997, [First author]

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## Invited Conference Presentations

1. “Atomic Hydrogen for Oxygen-Free Metal ALD and Bottom-Up Interconnect Fill,” ADMETA 2026, Japan, hybrid, October 7-9, 2026
2. “Current and Future Perspectives of Area-Selective Deposition,” ALD/ALE2026, Tampa, FL, USA, June 28, 2026
3. “Surface Chemistry to Device Innovation: Expanding the Frontiers of Atomic Layer Deposition,” 7th International Conference on Applied Surface Science, Chengdu, China, June 1, 2026
4. “Atomic Layer Deposition in the Age of AI: A Reinforcing Loop from Angstroms to Algorithms,” 2nd Chem & Bio Engineering Forum, Hangzhou, China, May 7, 2026
5. “Don’t Cover Everything: Why Semiconductor Processes Learned to Be Selective (다 덮지 마세요: 반도체 공정 이 선택을 배운 이유),” KALD 2026, Seoul, Republic of Korea, April 9, 2026
6. “New Paradigms in Atomic Layer Deposition for 3D Semiconductor Device Fabrication,” Japanese Society of Applied Physics Spring Meeting, Tokyo, Japan, March 15, 2026
7. “New Paradigms in Atomic Layer Deposition for 3D Semiconductor Device Fabrication,” RSC and EMRC Joint Symposium for Advancing Materials Science, Doha, Qatar, November 3, 2025
8. “Rediscovery of Atomic Layer Deposition to Overcome the Limitations of Semiconductor Manufacturing,” IMRC, Cancun, Mexico, August 18, 2025
9. “Mastering Selective Deposition in One Hour: Principles and Practice (선택적 증착 1시간만에 뽀개기 원리와 실 전),” KALD 2025, Seoul, Republic of Korea, April 24, 2025
10. “Building 3D AI Semiconductor Devices from the Ground Up: The Role of Atomic Layer Deposition (원자층 증 착법으로 부터 시작되는 3차원 AI 반도체 소자 공정기술),” AI Workshop, Korean Society of Electronic and Electrical Engineering, Seoul, Republic of Korea, April 17, 2025
11. “Area Selective Deposition Fundamentals & Applications, Semicon Korea, Seoul, Republic of Korea, February 19, 2025
12. “Rediscovery of ALD to Overcome the Limitations of Semiconductor Manufacturing,” Asian Pacific ALD 2024, Shanghai, China, October 17, 2024
13. “Technological Expansion of Atomic Layer Deposition to Patterning beyond Thin Film Deposition,” ACS 2024 Fall, Denver, CO, USA, August 19, 2024
14. “Area Selective Atomic Layer Deposition using a Size Cutter,” ALD/ALE2024, Helsinki, Finland, August 5, 2024

15. “Area Selective ALD using Homometallic Precursor Inhibitors for High Process Compatibility,” ASD Workshop 2024, Montreal, Canada, April 15, 2024
16. “Atomic Layer Deposition & Selective Growth for Nanofabrication of the Si Chips,” ICAE 2023, Jeju, Republic of Korea, November 2, 2023
17. “Area Selective Deposition using Homometallic Precursor Inhibitors,” ECS244, Gothenburg, Sweden, October 9, 2023
18. “Area Selective Deposition; The 4th Toolbox for Nanofabrication of Si Devices,” Nano Korea 2023, Goyang, Republic of Korea, July 6, 2023
19. “Computational Modeling of Physical Surface Reactions of Precursors in Atomic Layer Deposition by Monte Carlo Simulations,” APATCC-10, Quy Nhon, Vietnam, February 20, 2023
20. “Seam-less Deposition of $TiO_2$ using Gradient Selective Deposition,” KISM 2022, Pusan, Republic of Korea, November 4, 2022
21. “The 4th Toolbox for Semiconductor Device Nanofabrication beyond ALD,” Carbon Symposium, Seoul, Republic of Korea, September 23, 2022
22. “Another Opportunity in AS-ALD using Precursor Inhibitors,” ALD/ALE 2021, online, June 28, 2021
23. “Atomic Layer Deposition Fundamentals & Applications,” ICMAP, online, January 20, 2021
24. “Surface Energy Engineering by High & Low Technologies,” The 4th International Conference on Active Materials and Soft Mechatronics, Songdo, Republic of Korea, October 16 - 19, 2019
25. “Back to the Surface Science Again,” China ALD Workshop 2019, Huangshuan, China, August 8 - 9, 2019
26. “Surface-reactivity-determined Patterning Technology; The Era of Atomic Crafting,” ASD Workshop 2019, Leuven, Belgium, April 4 - 5, 2019
27. “Surface-reactivity-determined Patterning Technology; A Route to Atomic Craft,” Nano Convergence Conference 2019, Gangchon, Republic of Korea, January 17 - 18, 2019
28. “Surface Energy Tuning in Atomic Scale for Hydrophobic Coating,” The 4th International Conference on ALD Applications & 2018 China ALD Conference, Shenzen, China, October 14 - 17, 2018
29. “Atomic Layer Deposition of Rare Earth Oxides for Surface Energy Control,” The 7th International Conference on Microelectronics and Plasma Technology, Songdo, Republic of Korea, July 24 - 28, 2018

30. “Surface Functionalization for Conductivity Improvement by Metal Atomic Layer Deposition,” IEEE International Interconnect Technology Conference 2018, Santa Clara, CA, June 4 - 7, 2018
31. “Surface Energy Tuning in Atomic Scale for Hydrophobic Coating,” The Korean Ceramic Society Spring Meeting 2018, Changwon, Republic of Korea, April 11 - 13, 2018
32. “Atomic Layer Deposition on T-shirts,” American Chemical Society 2018 Spring National Meeting, New Orleans, LA, March 18 - 22, 2018
33. “Atomic Layer Deposited Metal Thin Films on T-shirts,” Korean Institute of Surface Engineering Meeting 2017, Jeju, Republic of Korea, November 22 - 25, 2017
34. “Atomic Layer Deposited Metal Thin Films on T-shirts,” Korean Vacuum Society Summer Meeting 2017, Hongcheon, Republic of Korea, August 16 - 18, 2017
35. “Defect Healing of Graphene for Conductive Flexible Electrodes by Wet & Dry Methods,” International Conference on Electronic Materials and Nanotechnology for Green Environment (ENGE) 2016, Jeju, Republic of Korea, November 6 - 9, 2016
36. “Atomic Layer Deposition of Metals on Textiles for Wearable Electronics,” China ALD 2016, Suzhou, China, October 16 - 19, 2016
37. “Electronic Textiles Fabricated using Atomic Layer Deposition,” CIMTEC 2016, Perugia, Italy, June 5 - 10, 2016
38. “Textile-Based Pressure Sensor Prepared by Atomic Layer Deposition,” Korean Society of Opticalelectronics Meeting 2016, Gwangju, Republic of Korea, January 26 - 27, 2016
39. “Atomic Layer Deposition of Rare Earth Oxide for Hydrophobic Coating,” Korean Institute of Surface Engineering Fall Meeting 2015, Yongin, Republic of Korea, November 26 - 27, 2015
40. “Opportunities of Atomic Layer Deposition for Electronic Textile Applications,” The Korean Fiber Society Fall Meeting 2015, Busan, Republic of Korea, November 5 - 6, 2015
41. “Opportunities in Atomic Layer Deposition for Electronic Textile and Hydrophobic Coating Applications,” 228th Electrochemical Society Meeting, Phoenix, AZ, USA, October 11 - 15, 2015
42. “ALD of Noble Metals for Energy Application,” 14th International Conference on Atomic Layer Deposition, Kyoto, Japan, June 15 - 18, 2014
43. “Pt Nanoparticle on Site-Constrained Surface by Atomic Layer Deposition,” The 6th Asian Conference on Crystal Growth and Crystal Technology, Jeju, Republic of Korea, June 11 - 14, 2014
44. “Pt Nanoparticle Fabrication by Atomic Layer Deposition with Self-Assembled Monolayers,” The Korean Ceramic Society Fall Meeting 2013, Jeju, Republic of Korea, October 16 - 18, 2013

45. “Nucleation Control in Atomic Layer Deposition by using Self-Assembled Monolayer,” Korean Materials Research Society Spring Meeting 2013, Yeosu, Republic of Korea, May 23-24, 2013
46. “Nucleation & Growth of ALD Metal on Carbon Surfaces,” Korean Atomic Layer Deposition Workshop 2013, Seoul, Republic of Korea, April 26, 2013
47. “Nucleation of 1D Pt Nanowires by Atomic Layer Deposition on Highly Ordered Pyrolytic Graphite,” 2013 Spring Conference of the Korean Institute of Metals & Materials, Jeju, Republic of Korea, April 25 - 26, 2013
48. “Atomic Layer Deposition of Metals on 2D Materials,” China Semiconductor Technology International Conference 2013, Shanghai, China, March 16 - 18, 2013
49. “Atomic Layer Deposition for Nanoscale Contact Applications,” International Conference on Electronic Materials and Nanotechnology for Green Environment, Jeju, Republic of Korea, November 21 - 24, 2010
50. “The Applications of Atomic Layer Deposition of Cobalt for Nanoscale Devices,” International Conference on Nano Science and Nano Technology 2009, Mokpo, Republic of Korea, November 5 - 6, 2009
51. “Atomic Layer Deposition for Nanoscale Contact Applications”, 25th International VLSI/ULSI Multilevel Interconnection Conference, Fremont, CA, USA, October 28 - 30, 2008
52. “Synchrotron X-ray Scattering Study on the Initial Growth of Atomic Layer Deposition Thin Films for the Next Generation MOSFET,” Electroceramics XI, Manchester, UK, August 31 - September 3, 2008

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## Invited Lectures

1. (Tutorial) “Atomic Layer Deposition and Atomic Layer Modeling: Surface Chemistry, Simulation, and Digital Twins for Semiconductor Processes,” ENGE 2026 (The 9th International Conference on Electronic Materials and Nanotechnology for Green Environment), The Korean Institute of Metals and Materials, Jeju Shinhwa World, Jeju, Republic of Korea, November 15, 2026
2. “AI-driven Digital Twin of ALD Process Environment: From Material Synthesis to Process Prediction,” AI + Ab Initio Modeling Forum, Applied Materials, online, August 21, 2026
3. “The One Who Directs AI, The One Who Thinks With AI: From Tool to Thinking Partner — How Asking Becomes a Skill,” VPP Seminar, Future Technology Research Institute, SK hynix, Icheon, Republic of Korea, August 19, 2026
4. “Teaching Atoms to Choose: Fundamentals and Applications of Area-Selective Atomic Layer Deposition,” Graduate School of Semiconductor Materials and Devices Engineering, UNIST, Ulsan, Republic of Korea, May 13, 2026
5. “Building the Future Angstrom by Angstrom: Atomic Layer Deposition Strategies for 3D Nanofabrication in the AI Era,” School of Chemical and Biological Engineering, Zhejiang University, Hangzhou, China, May 6, 2026
6. “A Experimental Learning Computational Approach: From Materials Design to Prediction of Molecular Behavior,” SK hynix, Icheon, Republic of Korea, April 8, 2026
7. “New Paradigms in Atomic Layer Deposition for 3D Semiconductor Device Fabrication,” Korea-Japan Joint Workshop for ALD, Kumamoto, Japan, February 24, 2026
8. “From Area-Selective Deposition to AI-Designed Materials: New Frontiers in Atomic Layer Deposition,” Korean Institute of Science and Technology, Seoul, Republic of Korea, February 12, 2026
9. “Atomic Layer Innovations: New Pathways for Future Semiconductor Fabrication,” Thin Film Society Webinar Series, Singapore, December 17, 2025
10. “New Paradigms in Atomic Layer Deposition for 3D Semiconductor Device Fabrication,” Department of Mechanical Engineering, Republic of Korea Advanced Institute of Science and Technology, Daejeon, Republic of Korea, December 15, 2025
11. “Atomic-Level Solutions for the Scaling Limits of Semiconductors,” SK TriChem, Sejong, Republic of Korea, November 12, 2025

12. “How to Effectively Utilize Large Language Models for Work and Research?” SK hynix, Icheon, Republic of Korea, October 17, 2025
13. “Expanding the Horizons of Atomic Layer Deposition for Next-Generation Semiconductor Fabrication,” TSMC, Taiwan, August 29, 2025
14. “Rediscovery of Atomic Layer Deposition to Overcome the Limitations of Semiconductor Manufacturing,” Department of Materials Science and Engineering, National Taiwan University, Taiwan, August 28, 2025
15. "Ending the 2nd Technological Generation of ASD & Moving Forward to the 3rd for High-Volume Manufacturing,” Semiconductor Laboratory, Samsung Electronics, Suwon, Republic of Korea, June 27, 2025
16. “Rediscovery of Atomic Layer Deposition to Overcome the Limitations of Semiconductor Manufacturing,” TEL Korea, Suwon, Republic of Korea, June 21, 2025
17. “The Industrial Ecosystem of Si Chips and Atomic Layer Deposition as a Key Nanofabrication Technology,” Department of Chemical Engineering, Wuhan University of Technology, Wuhan, China, May 16, 2025
18. “The Industrial Ecosystem of Si Chips and Atomic Layer Deposition as a Key Nanofabrication Technology,” Department of Mechanical Engineering, Huazhong University of Science and Technology, Wuhan, China, May 16, 2025
19. “Cracking Semiconductor Process Technology in One Hour (반도체공정기술1시간만에뽀개기),” Incheon POSCO High School, Incheon, Republic of Korea, May 13, 2025
20. “Rediscovery of Atomic Layer Deposition to Overcome the Limitations of Semiconductor Manufacturing,” Micron Technology, Boise, ID, USA, April 11, 2025
21. “Rediscovery of Atomic Layer Deposition to Overcome the Limitations of Semiconductor Manufacturing,” Intermolecular, Santa Clara, CA, USA, November 22, 2024
22. "The Industrial Ecosystem of Si Chips and Atomic Layer Deposition as a Key Nanofabrication Technology,” Department of Chemical Engineering, Stanford University, Stanford, CA, USA, November 20, 2024
23. "The Industrial Ecosystem of Si Chips and Atomic Layer Deposition as a Key Nanofabrication Technology,” Department of Chemical Engineering, Zhejiang University, Hangzhou, China, October 17, 2024
24. "The Industrial Ecosystem of Si Chips and Atomic Layer Deposition as a Key Nanofabrication Technology,” Department of Mechanical Engineering, POSTECH, Pohang, Republic of Korea, September 19, 2024

25. “Rediscovery of Atomic Layer Deposition to Overcome the Limitations of Semiconductor Manufacturing,” Department of Chemistry, University of Colorado Boulder, Boulder, CO, USA, August 20, 2024
26. “Supercritical Fluid Deposition for Semiconductor Device Fabrications,” Semiconductor Laboratory, Samsung Electronics, Suwon, Republic of Korea, July 6, 2024
27. “Rediscovery of Atomic Layer Deposition to Overcome the Limitations of Semiconductor Manufacturing,” Applied Materials, Santa Clara, CA, USA, May 17, 2024
28. “The 4th Toolbox for Semiconductor Device Nanofabrication beyond Atomic Layer Deposition,” Lawrence Livermore National Laboratory, Livermore, CA, USA, November 9, 2023
29. “The Industrial Ecosystem of Si Chips and Atomic Layer Deposition as a Key Nanofabrication Technology,” Department of Materials Science and Engineering, UNIST, Ulsan, Republic of Korea, October 25, 2023
30. “The Industrial Ecosystem of Si Chips and Atomic Layer Deposition as a Key Nanofabrication Technology,” ACS Science Talk, Online, September 8, 2023
31. “Why is My Advisor Always Upset for My Paper Figures?” Department of Materials Science and Engineering, Phenikaa University, Ha Noi, Vietnam, August 11, 2023
32. “The 4th Tool Box for Semiconductor Device Nanofabrication beyond Atomic Layer Deposition,” Department of Chemistry, University of Cologne, Cologne, Germany, June 3, 2023
33. “Area Selective Deposition; The 4th Toolbox for Si Device Nanofabrication,” Department of Materials Science and Engineering, KAIST, Deajeon, Republic of Korea, May 9, 2023
34. “The 4th Toolbox for Semiconductor Device Nanofabrication beyond ALD,” Department of Applied Physics, Kyunghee University, Suwon, Republic of Korea, April 25, 2023
35. “The 4th Toolbox for Semiconductor Device Nanofabrication beyond ALD,” Argonne National Laboratory, Chicago, IL, USA, March 29, 2023
36. “The 4th Toolbox for Nanofabrication of Si Devices,” Department of Materials Science and Engineering, Phenikaa University, Ha Noi, Vietnam, February 23, 2023
37. “Seam-less Deposition through Gradient Selective Deposition with Homometallic Precursor Inhibitors,” Semiconductor Laboratory, Samsung Electronics, Suwon, Republic of Korea, February 17, 2023
38. “Improvement of Selectivity in Area Selective Atomic Layer Deposition using a Size Cutter,” SK Hynix, Icheon, Republic of Korea, February 16, 2023

39. “Area Selective Deposition of Ru; A Key Process for Si Device Fabrication,” Tanaka Precious Metals, Tsukuba, Japan, January 31, 2023
40. “Icephobic Coating through Self-formed Superhydrophobic Layer,” Department of Materials Science and Engineering, January 5, 2023
41. “Area Selective Atomic Layer Deposition; The 4th Toolbox for Semiconductor Device Fabrication,” Department of Electrical and Electronic Engineering, Sungkyunkwan University, Suwon, Republic of Korea, January 3, 2023
42. “Area Selective Deposition; Past, Present, and Future,” Semiconductor Laboratory, Samsung Electronics, Suwon, December 1, 2022
43. “The 4th Tool Box for Semiconductor Device Nanofabrication beyond Atomic Layer Deposition,” Department of Chemistry, Carleton University, Ottawa, Canada, September 19, 2022
44. “The 4th Tool Box for Semiconductor Device Nanofabrication beyond Atomic Layer Deposition,” Department of Materials Science and Engineering, Hanyang University, Seoul, Republic of Korea, August 18, 2022
45. “How We Evaluate Scientific Papers?” SK hynix, Icheon, Republic of Korea, May 20, 2022
46. “Atomic Layer Deposition for Nanofabrication beyond Deposition,” Wuhan University of Technology, online, March 22, 2022
47. “The 4th Tool Box for Nanofabrication beyond ALD; Area Selective Deposition,” Applied Materials, online, December 9, 2021
48. “Recent Advances in ALD/ALE Technologies for Next-Generation Semiconductor Applications,” KETI, Sungnam, Republic of Korea, October 21, 2021
49. “The 4th Tool Box for Nanofabrication beyond Atomic Layer Deposition; Area Selective Deposition,” SK hynix, Icheon, Republic of Korea, October 19, 2021
50. “Application of Area Selective Deposition & Atomic Layer Modulation,” SK hynix, Icheon, Republic of Korea, October 1, 2021
51. “Area Selective Deposition And Applications,” Department of Materials Science and Engineering, Yeoungnam University, Gyeongsan, Republic of Korea, September 28, 2021
52. “Beyond ALD; Atomic Layer Modulation,” SK hynix, Icheon, Republic of Korea, August 24, 2021
53. “Beyond ALD; Atomic Layer Modulation,” SAIT, Samsung Electronics, Suwon, Republic of Korea, August 20, 2021
54. “How Can We Prepare Figures for Paper Writing?” Department of Materials Science and Engineering, Pusan National University, Pusan, Republic of Korea, July 12, 2021

55. “Fundamentals of Area Selective Deposition,” Semiconductor Laboratory, Samsung Electronics, Suwon, Republic of Korea, June 22, 2021
56. “Fundamentals of Area Selective Deposition,” SAIT, Samsung Electronics, Suwon, Republic of Korea, June 11, 2021
57. “The Era of Nanofabrication is Coming,” Department of Materials Science and Engineering, Seoul University of Science and Technology, Seoul, May 12, 2021
58. “Fundamentals of Area Selective Deposition,” SK hynix, Icheon, Republic of Korea, April 26, 2021
59. “Why is My Advisor Always Upset for My Paper Figures?” Department of Materials Science and Engineering, Yeoungnam University, Gyeongsan, Republic of Korea, April 19, 2021
60. “How Can We Prepare Figures for Paper Writing?” Department of Materials Science and Engineering, Hanyang University ERICA, Ansan, Republic of Korea, March 28, 2021
61. “The Era of Atomic Crafting: Area Selective Atomic Layer Deposition,” SK Materials, online, February 4, 2021
62. “Beyond ALD; Area Selective Deposition & Atomic Layer Modulation,” Department of Chemistry, Charleton University, online, December 15, 2020
63. “The Era of Atomic Crafting,” KETI, Sungnam, Republic of Korea, July 21, 2020
64. “Atomic Layer Deposition on Graphene,” Semiconductor Laboratory, Samsung Electronics, Suwon, Republic of Korea, July 13, 2020
65. “Science against Pseudoscience,” Department of Materials Science and Engineering, Hanyang University ERICA, Ansan, Republic of Korea, November 27, 2019
66. “The Era of Atomic Crafting; Area Selective Atomic Layer Deposition,” Department of Chemistry, National University of Singapore, Singapore, September 12, 2019
67. “Science against Pseudoscience,” Department of Materials Science and Engineering, Pusan National University, Pusan, Republic of Korea, September 5, 2019
68. “Surface-reactivity-determined Patterning Technology; The Era of Atomic Crafting,” Sungkyunkwan University, Suwon, Republic of Korea, August 12, 2019
69. “Surface-reactivity-determined Patterning Technology; The Era of Atomic Crafting,” University of California Merced, Merced, CA, USA, July 26, 2019
70. “Science against Pseudoscience,” Yeongnam University, Kyeongsan, Republic of Korea, July 17, 2019
71. “Atomic Layer Deposition; from Fundamentals to Applications,” Semiconductor Equipment Technology Center, Chunan, Republic of Korea, June 26, 2019

72. “Science against Pseudoscience,” Myungduk Girls’ Highschool, Seoul, Republic of Korea, March 22, 2019
73. “Atomic Layer Deposition; A Route to Atomic Craft,” Pusan National University, Pusan, Republic of Korea, November 20, 2018
74. “Atomic Layer Deposition; from Fundamentals to Applications,” Semiconductor Equipment Technology Center, Chunan, Republic of Korea, November 8, 2018
75. “Surface Energy Tuning in Atomic Scale for Hydrophobic Coating,” Department of Chemical Engineering, Hongik University, Seoul, November 11, 2018
76. “Atomic Layer Deposition; A Route to Atomic Craft,” Department of Mechanical Engineering, University of Hawaii Manoa, Honolulu, September 24, 2018
77. “A Surface-reactivity-determined Patterning Technology using Atomic Layer Deposition,” SK Hynix, Icheon, Republic of Korea, September 6, 2018
78. “Atomic Layer Deposition; A Route to Atomic Craft,” Instituto Tecnológico de Aeronáutica, São José dos Campos, Brazil, August 14, 2018
79. “From Surface Property Tuning to Real Applications,” Korea Institute of Science and Technology, Seoul, Republic of Korea, July 19, 2018
80. “Atomic Layer Deposition & High-k Dielectrics for Emerging Devices,” Samsung Advanced Institute of Technology, Suwon, Republic of Korea, July 11, 2018
81. “Thin Film Growth & Area Selective Atomic Layer Deposition,” Semiconductor Research Laboratory, Samsung Electronics, Suwon, Republic of Korea, June 18, 2018
82. “Thin Film Growth & Atomic Layer Deposition,” Samsung Electronics, Suwon, Republic of Korea, June 14, 2018
83. “Area-selective Atomic Layer Deposition; A Surface-reactivity-determined Patterning Technology,” Applied Materials, Santa Clara, CA, USA, June 7, 2018
84. “How Goals Can Make Our Life Better?” Myungduk Girls’ Highschool, Seoul, Republic of Korea, May 26, 2018
85. “Atomic Layer Deposition; from Fundamentals to Applications,” Semiconductor Equipment Technology Center, Chunan, Republic of Korea, May 3, 2018
86. “Atomic Layer Deposition; A Route to Atomic Craft,” Materials Chemistry Laboratory, Institute for Basic Science, Daejeon, Republic of Korea, May 1, 2018

87. "Hydrophobic Coating by High Tech and Low Tech Approaches," Department of Materials Science and Engineering, Yeongnam University, Kyeongsan, Republic of Korea, December 19, 2017
88. "From Surface Property Tuning to Real Applications," Department of Materials Science and Engineering, Kyung Hee University, Suwon, Republic of Korea, November 11, 2017
89. "Atomic Layer Deposition, A Key Player in Semiconductor Technology," Lam Research, Fremont, CA, USA, July 20, 2017
90. "From Surface Property Tuning to Real Applications," Department of Nano and Energy Engineering, Pusan National University, Pusan, Republic of Korea, June 16, 2017
91. "Atomic Layer Deposition; from Fundamentals to Applications," Semiconductor Equipment Technology Center, Chunan, Republic of Korea, May 17, 2017
92. "How Nanotechnologies Can Make Our Life Better?," Myungduk Highschool, Seoul, Republic of Korea, April 21, 2017
93. "Electronic Textiles fabricated by Atomic Layer Deposition for Wearable Electronics," Korea Institute of Industrial Technology (KITECH), Ansan, Republic of Korea, February 28, 2017
94. "Electronic Textiles Fabricated by Atomic Layer Deposition for Wearable Electronics and Bio Applications," Department of Chemical Engineering, Chungnam National University, Daejeon, Republic of Korea, February 23, 2017
95. "Surface Functionalization of Graphene by Atomic Layer Deposition," Electronic Convergence Material and Device Research Center, Republic of Korea Electronics Technology Institute (KETI), Sungnam, Republic of Korea, December 5, 2016
96. "Atomic Layer Deposition Beyond High Technology; Electronic Textiles, Hydrophobic Filter, and Transparent 2D Heater," Electronic Convergence Material and Device Research Center, Republic of Korea Electronics Technology Institute (KETI), Sungnam, Republic of Korea, November 11, 2016
97. "Atomic Layer Deposition Beyond High Technology; Electronic Textiles, Hydrophobic Filter, and Transparent 2D Heater," School of Energy and Chemical Engineering, Ulsan National Institute of Science and Technology (UNIST), Ulsan, November 9, 2016
98. "Atomic Layer Deposition; from Fundamentals to Applications," Semiconductor Equipment Technology Center, Chunan, Republic of Korea, October 11, 2016
99. "Atomic Layer Deposition; An Essential Tool for Nanofabrication," Department of Chemistry, Thammasat University, Bangkok, Thailand, October 6, 2016

100. “Atomic Layer Deposition; An Essential Tool for Nanofabrication,” Department of Materials Science and Engineering, Kasetsart University, Bangkok, Thailand, October 5, 2016
101. “Surface Functionalization for Property Tuning,” Division of Materials Science and Engineering, Hanyang University, Seoul, Republic of Korea, September 27, 2016
102. “Conductive Textile for Sensors and Heaters,” School of Materials Science and Engineering, Yongnam University, Gyeongsan, Republic of Korea, September 21, 2016
103. “Atomic Layer Deposition; An Essential Tool for Nanofabrication,” Tera Semicon, Anseong, Gyeonggi, Republic of Korea, September 9, 2016
104. “Atomic Layer Deposition; An Essential Tool for Nanotechnology,” Department of Materials Science and Engineering, Sejong University, Seoul, Republic of Korea, June 16, 2016
105. “Atomic Layer Deposition; from Fundamentals to Applications,” Semiconductor Equipment Technology Center, Chunan, Republic of Korea, May 20, 2016
106. “Surface Functionalization for Property Tuning,” School of Advanced Materials Engineering, Sungkyunkwan University, Suwon, Republic of Korea, May 12, 2016
107. “Surface Functionalization for Property Tuning,” Department of Mechanical Engineering, Kyunghee University, Suwon, Republic of Korea, May 9, 2016
108. “Defect Healing of Graphene,” School of Materials Science and Engineering, Yongnam University, Gyeongsan, Republic of Korea, April 1, 2016
109. “Platinum Atomic Layer Deposition,” Department of Mechanical Engineering, Pusan National University, Yongin, Republic of Korea, December 29, 2015
110. “Atomic Layer Deposition for Nanofabrication,” Myungji University, Yongin, Republic of Korea, November 6, 2015
111. “Atomic Layer Deposition; from Fundamentals to Applications,” Semiconductor Equipment Technology Center, Chunan, Republic of Korea, October 5, 2015
112. “Atomic Layer Deposition; An Essential Tool for Nanotechnology,” Department of Energy Science, Sungkyunkwan University, Suwon, Republic of Korea, September 30, 2015
113. “Tutorial: Atomic Layer Deposition,” Advanced Metallization Conference 2015, Seoul, Republic of Korea, September 16, 2015
114. “Atomic Layer Deposition; from Fundamentals to Applications,” Semiconductor Equipment Technology Center, Chunan, Republic of Korea, May 8, 2015
115. “Applications of Atomic Layer Deposition Pt,” Tanaka Research Center, Tsukuba, Japan, April 2, 2015

116. “Series Class of Atomic Layer Deposition,” Samsung Electronics, Hwasung, Republic of Korea, March 5-6, 2015
117. “Research Topics of the Lee’s Group,” Wonik IPS, Pyeongtaek, Republic of Korea, March 2, 2015
118. “Material Synthesis and Applications by Atomic Layer Deposition,” Graduate School of Convergence Science and Technology, Seoul National University, Seoul, Republic of Korea, January 19, 2015
119. “Low Temperature Atomic Layer Deposition of Pt for Electronic Textile Applications,” State Key Laboratory, Nanjing Tech University, Nanjing, China, January 16, 2015
120. “Surface Functionalization by Atomic Layer Deposition,” Samsung Medical Center, Seoul, Republic of Korea, November 26, 2014
121. “Atomic Layer Deposition,” e-Nanoschool 2014 by Korea Nano Technology Research Society, 6 on-line classes from November 18 to December 4, 2014
122. “Atomic Layer Deposition on 2D Materials,” School of Energy and Chemical Engineering, Ulsan National Institute of Science and Technology (UNIST), Ulsan, November 5, 2014
123. “Atomic Layer Deposition; from Fundamentals to Applications,” Semiconductor Equipment Technology Center, Chunan, Republic of Korea, October 8, 2014
124. “Atomic Layer Deposition of Pt and Applications,” Shonan Site, Tanaka Kikinzoku Kogyo K.K., Hiratsuka, Japan, September 19, 2014
125. “Atomic Layer Deposition of Pt/TiO2 for VOC Removal,” Head Office, Tanaka Kikinzoku Kogyo K.K., Tokyo, Japan, September 18, 2014
126. “Atomic Layer Deposition of Metals on 2D Materials and Applications,” Department of Materials Science and Engineering, Hanyang University, Seoul, Republic of Korea, July 30, 2014
127. “Pt/Carbon by Atomic Layer Deposition for Catalyst, Electrode, and Sensor Applications,” Electronic Materials Research Center, Republic of Korea Institute of Science and Technology (KIST), Seoul, Republic of Korea, July 28, 2014
128. “Atomic Layer Deposition; from Fundamentals to Applications,” Semiconductor Equipment Technology Center, Chunan, Republic of Korea, May 15, 2014
129. “Atomic Layer Deposition for Emerging Applications,” Department of Polymer Science and Engineering, Korean National University of Transportation, Chungju, Republic of Korea, December 23, 2013
130. “Atomic Layer Deposition for Interconnect Technology,” Inter-University Semiconductor Research Center, Seoul National University, Seoul, Republic of Korea, November 29, 2013

131. “What is a Scientific Paper?,” School of Materials Science and Engineering, Pusan National University, Pusan, Republic of Korea, November 7, 2013
132. “Atomic Layer Deposition of Pt on Carbon Surfaces; Graphite and Graphene,” Korea Institute of Machinery and Materials, Daejun, Republic of Korea, September 26, 2013
133. “Atomic Layer Deposition of Pt on Carbon Surfaces; Graphite and Graphene,” Korea Research Institute of Standards and Science, Daejun, Republic of Korea, September 26, 2013
134. “Atomic Layer Deposition of Pt on Carbon Surfaces,” Department of Mechanical Engineering, Republic of Korea University, Seoul, Republic of Korea, August 20, 2013
135. “Atomic Layer Deposition of Pt on Carbon Surfaces” Department of Materials Science and Engineering, Yeongnam University, Daegu, Republic of Korea, June 26, 2013
136. “Atomic Layer Deposition of Pt on Carbon Surfaces,” Graduate School of Energy, Environment, Water, and Sustainability, Republic of Korea Advanced Institute of Science and Technology, Daejeon, Republic of Korea, May 8, 2013
137. “Atomic Layer Deposition for Nanotechnologies,” Department of Chemical Engineering, Inha University, Incheon, Republic of Korea, May 3, 2013
138. “Nucleation & Growth of ALD Metal on Carbon Surfaces,” Korean ALD Workshop 2013, Seoul, Republic of Korea, April 26, 2013
139. “How Nanotechnologies Can Make Our Life Better?,” Myungduk Highschool, Seoul, Republic of Korea, April 23, 2013

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## Professional Activities

### CURRENT APPOINTMENTS

- Executive Editor of Chemistry of Materials, ACS Publications, 2025 - present
- Visiting Consulting Professor, SK hynix, 2026 - present
- Program Committee, ALD/ALE, 2018 - present
- Steering Committee, ALD, 2026 - present
- Program Committee, AVS, 2025 - present
- Program Committee, Euro CVD, 2025 - present
- International Committee, Asian Pacific ALD, 2018 - present
- Editorial Board Member, 2015 - present, Electronic Materials Letters (http://link.springer.com/journal/13391)

### PREVIOUS APPOINTMENTS

- Associate Editor of Chemistry of Materials, ACS Publications, 2018 - 2025
- Chair, International Conference ALD/ALE 2025, 2025
- Visiting Consulting Professor, SK hynix, 2021-2022
- Chair, Workshop on Area Selective Deposition, 2023
- Co-organizer, ACS Award in Surface Chemistry: Symposium in honor of Stacey F. Bent, 255th ACS National Meeting, 2018, New Orleans, LA
- Scientific Program Committee, 7th International Conference on Microelectronics and Plasma Technology (ICMAP 2018), 2018, Republic of Korea
- Guest Editor for ACS Virtual Issue, “Recent Advances in Atomic Layer Deposition,” 2016 (http://pubs.acs.org/page/vi/advances-ald.html)
- Organizing committee, Advanced Metallization Conference 2015 (ADMETA 2015), 2015, Republic of Korea
- General Secretary, Devision of Materials Science Convergence in The Korean Institute of Metals and Materials, 2015, Republic of Korea
- Organizer, 2nd Korean Meeting on Atomic Layer Deposition for Emerging Technologies, supported by The Korean Information Display Society, 2014, Republic of Korea
- Advisory Committee, Devision of Nanoprocessing & Nanoanalysis Tools, Selection of Nano R&BD Topics, 2014, Republic of Korea
- Symposium Organizer, Division of Equipments and Tools for Display Technology, The 14th International Meeting on Information Display, 2014, Republic of Korea

- Symposium Organizer and Chair, “Atomic Layer Deposition for Energy Applications,” Korean Ceramic Society Fall Meeting 2013, 2013, Republic of Korea
- Organizer, 1st Korean Meeting on Atomic Layer Deposition for Emerging Technologies, supported by The Korean Information Display Society, 2013, Republic of Korea
- Advisory Committee, Devision of Nanoprocessing & Nanoanalysis Tools, Nano Roadmap 2nd Edition, 2013, Republic of Korea

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## Fellowship and Honor

- Reseach and Science Award of Incheon Metropolitan City 2017
  - Date: 4/9/2018
- Young Scientist Award, The Korean Institute of Metals and Materials 2015
  - Date: 4/23/2015
- Best Paper Award, Material Research Society of Korea 2009
  - Title: High quality nickel atomic layer deposition for nanoscale contact applications
  - Date: 5/21/2009
- Graduate Student Award (Outstanding Paper), Dept. of MSE, POSTECH
  - Title: Spontaneous formation of vertical magnetic metal nanorod arrays during plasma enhanced atomic layer deposition
  - Date: 12/15/2008
- Samsung Award (Best Paper Award), The 15th Korean Conference on Semiconductor
  - Title: High quality epitaxial $CoSi_2$ using plasma nitridation-mediated epitaxy
  - Date: 2/22/2008
- Graduate Student Award (Outstanding Conf. Presentation), Dept. of MSE, POSTECH
  - Title: High quality epitaxial $CoSi_2$ using plasma nitridation-mediated epitaxy
  - Date: 12/15/2007
- Best Poster Award, Material Research Society 2007 Fall Meeting
  - Title: Analytical study on initial growth stage of metal atomic layer deposition by synchrotron radiation X-ray reflectivity analysis
  - Data: 11/28/2007
- University fellowship, Sungkyunkwan University
  - Period: 2002 - 2004

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## Teaching Activities

- MSE Laboratory 1
  - For sophomore, spring
- Modern Physics
  - For sophomore, fall
- Semiconductori Device Nanofabrications 1
  - For junior, spring
- Semiconductori Device Nanofabrications 2
  - For junior, fall
- Materials Analysis
  - For graduate student
- Materials Kinetics
  - For graduate student
- Thin Film Deposition
  - For graduate student

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## University and Departmental Committees

- Advisory Committee, INU Center for Research Facilities, 2014 – 2019
- Committee, Task Force Team of College of Engineering, INU, 2015 – 2019
- Committee, Graduate School of INU, 2018 – 2019
- Committee, Global Research Track, INU, 2019 – 2020
- Department Chair, Department of Materials Science and Engineering, INU, 2022 – 2023

## Press

1. “Electronic Textiles fabricated by Low Temperature Atomic Layer Deposition,” in The Kukmin Daily, Yonhap News, Kyeonggi Daily, Herald Economy, NSP News Agency, Maeil Daily, Daehak News, Bridge News,12/5/2016.
2. “Defect Healing of Graphene by Galvanic Displacement Reaction for Transparent Electrode,” in YP News, The Electronic Times, Veritas Alpha, Energy & Economy Times, 4/20/2016.
3. “Guest Editor for ACS Virtual Issue (Recent Advances in Atomic Layer Deposition),” in Aju Business Daily, Gukje News, Sport Kyunghyang, Bridge News, Asia Today, Kyeonggi Daily, Kyeongin Daily, 4/14/2016.
4. "INU Department of Materials Science Engineering and Samsung Advanced Institute of Technology Publish Nano-Process Research in Nature Communications," in Daihan Economy, Sports Kyunghyang, eNews Today, Kiho Ilbo, 12/13/2022. [Link]
5. "Prof. Han-Bo-Ram Lee (INU) and Prof. Il-Kwon Oh (Ajou University) Transfer KRW 1.175 Billion in Semiconductor Research Equipment Technology," in Daihan Economy, Seoul Ilbo, Gyeonggi Domin Ilbo, INU News, 12/20/2023. [Link]
6. "INU Team Publishes Ultra-Thin Insulating Film Technology for Next-Generation Semiconductors in ACS Nano," in Naewoeilbo, Jeonmae, Asia Ilbo, 2025. [Link]
7. "INU Research Team's ALD Equipment and Process Technology Transferred to CN1, Opening a New Horizon in AI Computing Speed," in Naewoeilbo, Bridge Economy (Viva100), INU News, 9/25/2025. [Link 1] [Link 2]
8. "INU Prof. Han-Bo-Ram Lee Research Team Publishes Next-Generation Semiconductor Ruthenium Interconnect Technology in JACS," in Gyeongin Newspaper, Gukje News, Newstown, Gyeonginmaeil, Korean Media News, Newsis, Golden Times, 6/10/2026. [Link]

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